Project Report

Project Phase-I

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ELECTRONICS AND COMMUNICATIONS ENGINEERING (ECE)

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled "**WIRELESS CHARGING FOR ELECTRIC VEHICLES"** submitted submitted by

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in partial fulfillment of the requirements for the award of the Degree Bachelor of Technology in Electronics and Communication Engineering is a bonafide record of the work carried out under my guidance and supervision at Amrita School of Engineering, Bengaluru.

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ABSTRACT

The rapid adoption of Electric Vehicles (EVs) is a defining milestone in the transition toward sustainable and low-carbon transportation. However, present-day EVs depend heavily on plug-in charging, which poses several usability and safety concerns such as manual cable handling, connector deterioration over repeated use, and vulnerability to water and environmental exposure. These limitations highlight the need for safer, more autonomous, and user-friendly charging alternatives. To overcome these challenges, this project proposes a wireless Electric Vehicle charging system based on resonant magnetic power transfer, enabling contactless energy delivery without the use of conventional charging cables.

The proposed solution employs resonant inductive coupling between a ground-embedded transmitter and a vehicle-mounted receiver coil, designed to achieve efficient power transfer over practical air-gaps. The system incorporates inverter-based high-frequency power conversion, compensation networks for resonance tuning, buck-based regulation for battery charging, and an automated vehicle-detection mechanism for intelligent power activation. A detailed simulation model was developed in MATLAB/Simulink to analyze coil resonance, voltage/current waveforms, misalignment sensitivity, charging behavior, and converter response under varying load conditions. To validate the simulation, a small-scale hardware prototype was constructed using custom-wound coils, an Arduino-controlled relay system, IR-based vehicle presence sensing, and visual charging indication through onboard LEDs and an LCD status display.

Simulation and prototype results confirm successful wireless energy transfer and demonstrate how automated sensing and switching improve operational convenience and safety. This work serves as a foundational framework for scalable wireless EV charging and lays the groundwork for future enhancements including higher-power operation, real-time communication for billing and control, intelligent alignment, and dynamic charging while in motion. The project aligns strongly with Sustainable Development Goals (SDG-7: Affordable & Clean Energy, SDG-9: Industry, Innovation & Infrastructure), reinforcing its relevance for next-generation smart mobility ecosystems.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURES

- **EV**: Electric Vehicle
- **WPT**: Wireless Power Transfer
- **ECE**: Electronics and Communication Engineering
- **SDG**: Sustainable Development Goal
- **IEEE**: Institute of Electrical and Electronics Engineers
- **EMI**: Electromagnetic Interference
- **KPI**: Key Performance Indicator
- **IR**: Infrared
- **LCD**: Liquid Crystal Display
- **BLDC**: Brushless DC Motor
- **AC**: Alternating Current
- **DC**: Direct Current

CHAPTER 1: INTRODUCTION

The global shift towards sustainable transportation has accelerated the adoption of Electric Vehicles (EVs). While EVs offer significant environmental benefits by reducing greenhouse gas emissions and reliance on fossil fuels, the practicality of their charging infrastructure remains a key barrier to widespread adoption. This project focuses on overcoming the limitations of conventional charging methods through the development of a wireless charging system.

1.1 Overall System/Scenario

The current EV charging ecosystem is dominated by conductive (plug-in) charging stations. Despite their rapid proliferation, these traditional systems present several user challenges. Users must manually handle bulky, heavy cables, which can be difficult for some individuals and inconvenient in adverse weather conditions like rain or snow. The physical connectors and cables are also susceptible to significant wear and tear from repeated use, as well as damage from environmental factors and vandalism. This leads to high maintenance costs and potential reliability issues. Furthermore, the reliance on high-power physical connections creates inherent safety risks, including tripping hazards from cables and the potential for electrical exposure or short-circuiting. These inconveniences and hazards detract from a seamless user experience and can dissuade potential EV buyers.

To address these limitations, this project proposes a wireless charging solution based on resonant magnetic coupling. This technology, also known as Wireless Power Transfer (WPT), offers a paradigm shift by eliminating the physical connection. Power is transferred wirelessly from a charging pad embedded in the ground (transmitter) to a receiver coil integrated into the undercarriage of the vehicle. When the vehicle is parked over the pad, an alternating magnetic field is generated by the transmitter, which induces a current in the receiver coil. This current is then rectified and used to charge the vehicle's battery.

This approach offers enhanced safety by eliminating exposed electrical contacts, reduces wear and tear, and provides unparalleled convenience—the user simply parks the car to initiate charging. As efficiency levels of WPT systems begin to approach those of wired systems [1], wireless charging is poised to be a key enabler for the widespread adoption of EVs and the development of smart, sustainable urban environments, including future applications like dynamic charging for autonomous fleets [2, 3].

1.2 Objective

The overarching aim of this project is to conceptualize, design, and demonstrate a functional wireless charging system tailored for electric vehicles. As the world steadily shifts toward clean and sustainable mobility, the demand for safer, more intuitive, and automated charging technologies is rapidly increasing. This project contributes to that vision by developing a system that replaces cumbersome plug-in chargers with a convenient, contactless power transfer mechanism, offering users a seamless charging experience.

Primary Objective

To develop a complete wireless charging system capable of reliably and efficiently transferring electrical power to an electric vehicle without the use of physical charging cables, thereby

improving convenience, safety, and long-term usability.
This objective emphasizes not only the technical performance of the system but also its user-centered nature. A properly designed wireless charging platform eliminates handling of cables, prevents connector wear-and-tear, reduces electrical exposure, and makes EV charging more intuitive—similar to parking a car and letting the system do the rest.

Sub-Objectives

1. Transmitter Design
To design, model, and optimize the primary side of the wireless charging system, which is responsible for generating the electromagnetic field used for power transfer.
This includes:

- Designing a transmitter coil with appropriate geometry, wire gauge, number of turns, and dimensions to create a stable and sufficiently strong magnetic field.
- Developing a high-frequency inverter circuit capable of energizing the transmitter coil at the required resonant frequency, which is typically in the range of tens of kilohertz.
- Implementing an appropriate compensation network (such as Series–Series or LCC) to maximize power transfer efficiency, reduce reactive power flow, and ensure resonance is maintained under different loading conditions.
- Ensuring electrical safety and thermal stability, especially during prolonged operation or high-frequency switching.

This objective ensures that the transmitter unit forms a robust foundation capable of delivering reliable wireless energy.

2. Receiver Design
To create a receiver subsystem capable of efficiently capturing the transmitted magnetic energy and converting it into a clean, stable DC output suitable for charging the EV battery.
This involves:

- Designing a receiver coil that is well-matched to the transmitter in terms of inductance, resonance frequency, and alignment tolerance.
- Developing a rectifier and filtering stage that converts the received AC waveform into smooth DC voltage with minimal ripple.
- Ensuring voltage and current levels remain within safe operating limits for the battery, thereby preventing thermal or electrical stress.
- Testing the receiver's performance under various distances and misalignment conditions to assess real-world feasibility.

The receiver plays a crucial role in ensuring that the delivered wireless power is usable, efficient, and stable.

3. Hardware Implementation
To translate the simulated system into a small-scale, fully functional prototype that demonstrates real-world feasibility and user interaction.
This objective includes:

- Constructing a physical transmitter pad with the designed coil and integrating it with a driver circuit and compensation network.
- Embedding a miniaturized receiver coil into a model EV, along with LEDs or a small DC load for visual confirmation of received power.

- Implementing vehicle detection and automation using IR sensors, relays, and an Arduino microcontroller to mimic the logic of real EV charging stations.
- Providing user feedback through an LCD display, status indicators, and automated power control.
- Evaluating how the system responds in different practical scenarios—distance variations, coil misalignment, sensor activation delays, and energy transfer consistency.

The objective ensures that the prototype not only demonstrates wireless charging in action but also mirrors the functionality, safety, and automation of commercial systems.

1.3 Methodology

The project adopts a structured and systematic methodology that blends theoretical research, simulation-driven design, and practical hardware development. This hybrid approach ensures that the final prototype not only works in theory but also performs reliably in real-world conditions. By combining MATLAB/Simulink modeling with hands-on experimentation, the project balances engineering rigor with practical usability.

1. Literature Survey

The process begins with an extensive literature review covering wireless power transfer technologies, resonant inductive coupling, compensation topologies, coil design strategies, and power-electronic drivers. Understanding past research helps identify which methods yield higher efficiency, which compensation networks perform better under misalignment, and what limitations current systems face. This step ensures that the project is built on established engineering principles rather than trial-and-error, allowing the design choices to be grounded in proven methods.

2. System Design

Using the insights gained from literature, the overall system architecture is developed. This involves defining the roles of the transmitter, receiver, inverter, compensation network, and sensing blocks. Each subsystem is designed with clear objectives—such as generating a strong magnetic field on the transmitter side or producing a stable DC output on the receiver side. Control logic is also incorporated at this stage, detailing how the system should detect vehicles, when power should be transmitted, and how feedback should be provided to the user. This architectural blueprint acts as the foundation for both simulation and hardware implementation.

3. Simulation

Before constructing the hardware, the entire wireless charging system is modeled in MATLAB/Simulink. This virtual environment allows testing of high-frequency inverter behavior, coil coupling efficiency, compensation tuning, voltage regulation, and system stability—without risking physical components. Simulations are run under various conditions, including coil misalignment, air-gap variations, and load changes, to understand how the system behaves in real-world scenarios. By analyzing waveform behavior, power transfer efficiency, current flow, and resonance response, the design can be fine-tuned to achieve optimal performance before moving to hardware.

4. Prototyping

Once the simulation results confirm that the system behaves as expected, a scaled hardware prototype is built. This prototype includes custom-wound transmitter and receiver coils, IR-based vehicle detection, relay switching, and Arduino-based control logic. Components such as buck converters, rectifiers, and LCD modules are added to replicate real charging conditions. The goal of the prototype is not to achieve high power, but to demonstrate the functional behavior of wireless charging, sensing automation, and system-level operation. This hands-on stage transforms abstract simulation data into a tangible working system.

5. Testing & Analysis

The final stage involves detailed testing and performance evaluation of both the simulation model and the physical prototype. Measurements such as coil alignment sensitivity, wireless power transfer effectiveness, LED brightness stability, sensor responsiveness, and relay switching accuracy are recorded. The behavior of the system is compared against simulation results to verify consistency between theory and practice. Any deviations provide valuable insights into hardware limitations, parasitic losses, or control timing issues. This analysis helps refine the design and identifies improvements for future iterations, such as higher power levels, better shielding, or more advanced compensation networks.

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1.4 Expected Result/Obtained Result

Anticipated Outcomes:

- A functional, detailed simulation model in MATLAB/Simulink that validates the wireless power transfer efficiency and stability of the designed system.
- A small-scale hardware prototype that successfully demonstrates the core functionalities:
  - Automatic detection of a vehicle's presence and alignment.
  - User feedback via an LCD screen.
  - Successful wireless transfer of power to the model vehicle (e.g., lighting an LED).

Obtained Results (Phase 1):

- A comprehensive literature survey has been completed.
- The end-to-end simulation model has been designed and integrated.
- A functional hardware prototype has been built and successfully demonstrates vehicle detection using IR sensors, user feedback on an LCD, and wireless power transfer to a model vehicle, as shown in the demonstration chapters.

1.5 Time table

The Phase 1 of this project followed a structured four-month timeline.

Table 1.1: Project Phase 1 Timeline

Month (Phase 1)	Key Tasks and Deliverables	Status
August	**Literature Survey:** Comprehensive review of existing research, technologies, and challenges in wireless EV charging.	✅ Completed
September	**Simulation of Transmitter Circuit:** Detailed design and simulation of the primary coil and associated power electronics.	✅ Completed
October	**Simulation of Receiver Circuit & Integration:** Design and simulation of the secondary coil, rectifier, and battery management interface. Full system integration.	✅ Completed
November	**Hardware Implementation & Final Evaluation:** Assembly of the hardware prototype, testing, performance evaluation, and final report compilation.	✅ Completed

1.6 Contents of rest of the report

This report is structured into six chapters.

- **Chapter 2** presents the "State of Art," reviewing the foundational background of wireless power transfer and existing literature in the field.
- **Chapter 3** details the "Design and Development," covering the system architecture, simulation model, and hardware component selection.
- **Chapter 4** describes the "Implementation and Demonstration," explaining the construction of the hardware prototype.
- **Chapter 5** provides the "Results and Discussion" from both the simulation and the hardware prototype.
- **Chapter 6** concludes the report with a summary of the findings and outlines the "Conclusion and Future Scope."

CHAPTER 2: STATE OF ART

2.1 Introduction

This chapter presents a comprehensive overview of the technological landscape surrounding wireless charging for electric vehicles. It begins by introducing the fundamental concepts of Wireless Power Transfer (WPT), highlighting how magnetic coupling and resonant compensation make contactless energy transfer possible. The discussion then progresses into a detailed examination of the existing research and developments in the field, exploring various coil designs, compensation networks, power-electronic methods, and alignment-assistance techniques proposed by different researchers. By reviewing advancements as well as the limitations reported in previous studies, this chapter establishes a clear understanding of current capabilities, ongoing challenges, and the motivation for further innovation in wireless EV charging systems.

2.2 Basic Background of Wireless Power Transfer (WPT)

Wireless Power Transfer (WPT) refers to the delivery of electrical energy from a source to a load without the need for physical wires, enabling safe and convenient power delivery in many modern applications. Among the various WPT techniques, magnetic resonant coupling has emerged as the most practical and efficient approach for high-power electric vehicle (EV) charging. In this method, both the transmitter and receiver coils are tuned to operate at the same resonant frequency, which allows them to exchange energy efficiently through an alternating magnetic field generated at high frequency. When resonance is achieved, the system behaves like a highly selective energy channel, significantly improving power transfer efficiency even when there is physical distance or slight misalignment between the coils, making it ideal for EV applications (Ahn & Park, 2018; Li & Mi, 2015).

In a typical resonant inductive charging system, the transmitter coil is energized using a high-frequency inverter, which converts the DC supply into a rapidly oscillating AC current. This AC current produces the time-varying magnetic field required for wireless energy transfer. To ensure maximum power delivery, both sides of the system incorporate compensation networks—such as Series–Series (S–S), Series–Parallel (S–P), or LCC topologies—which help achieve resonance, reduce reactive power, and maintain a high power factor (Villa et al., 2012). The receiver coil captures the magnetic energy, converts it back into electrical energy using a rectifier, and regulates the output using a DC–DC converter or charge controller to safely charge the EV battery (Liu et al., 2018).

Magnetic resonant coupling is especially beneficial because it supports medium-range power transfer relative to coil size, provides higher tolerance to coil misalignment, and can achieve relatively high efficiency when properly tuned. This is a major improvement over traditional inductive coupling, which suffers from efficiency loss whenever alignment is not perfect. Advanced coil geometries, ferrite structures, and shielding materials are often used to improve coupling strength and reduce leakage flux in practical EV chargers (Budhia et al., 2011).

Moreover, modern systems increasingly integrate communication, alignment detection, and intelligent control, enabling features like automatic activation, billing, and safety shutdown (RamRakhyani et al., 2021).

Overall, magnetic resonant coupling forms the foundation of today's research in wireless EV charging and continues to be explored for its potential in static charging stations, dynamic charging lanes, and smart transportation ecosystems. The combination of high efficiency, reasonable tolerance to misalignment, and compatibility with real driving environments makes it one of the most promising technologies for future mobility infrastructure.

2.3 Related Literature in Detail

A survey of existing literature reveals several key areas of research. The findings from a few foundational papers are summarized in Table 2.1.

Table 2.1: Summary of Literature Survey

Author(s), Year	Title	Focus	Key Findings
A. K. Singh et al., 2023 [1]	Wireless Charging in a Dynamic Environment for EVs	Dynamic alignment wireless charging	Demonstrates that stable power transfer is possible even with vehicle movement. However, efficiency is shown to decrease significantly with coil misalignment.
A. K. RamRakhyani et al., 2021 [2]	Simultaneous Wireless Power and Data Transfer	Power + data transfer integration	Explores the integration of data communication (e.g., billing, battery status) with power transfer. This improves automation but highlights challenges in EMI interference and efficiency trade-offs.
M. S. Islam et al., 2022 [3]	Efficiency Enhancement of Wireless Charging Systems	Techniques to boost charging efficiency	Provides a comprehensive review of methods to improve efficiency, such as advanced coil design and compensation networks. It notes that thermal losses remain a significant challenge.

CHAPTER 3: DESIGN AND DEVELOPMENT

3.1 Introduction

This chapter focuses on the complete design and analytical development of the wireless electric vehicle charging system, bridging the gap between theoretical understanding and practical implementation. It outlines the systematic process used to build the system—from conceptual modeling to hardware realization—and explains how each design choice contributes to achieving efficient and reliable wireless power transfer. The chapter begins by presenting the overall system architecture, highlighting how the transmitter, receiver, compensation network, and control logic interact as a unified system. It then explores the MATLAB/Simulink simulation environment, where key parameters such as resonant frequency, coupling coefficient, coil alignment, and power flow were analyzed to validate system behavior before hardware construction. Finally, the chapter details the component selection, coil design, sensing mechanisms, and control circuitry used in the physical prototype. Together, these sections provide a comprehensive understanding of how the system was engineered, optimized, and prepared for real-world testing.

3.2 System Architecture

The wireless EV charging system is organized around two major subsystems that work together to deliver safe, automated, and efficient wireless power transfer:
(1) the ground-based Charging Station (transmitter side) and
(2) the vehicle-mounted EV System (receiver side).
This architectural division reflects how real-world wireless charging systems are deployed—where the charging pad integrated into the parking surface supplies power, and the EV carries the receiving hardware. The overall architecture ensures smooth coordination between these units through sensing, control logic, and resonant power transfer.
The transmitter side handles all power generation and vehicle detection functions. It includes the high-frequency inverter, compensation network, and transmitter coil responsible for producing the alternating magnetic field. A microcontroller-based control unit continuously monitors the IR sensors placed near the charging pad. When a vehicle is detected, the control logic activates a relay system that energizes the transmitter coil. This prevents unnecessary power transmission and improves safety by ensuring the charger operates only when an EV is properly positioned.
The receiver side, mounted beneath the vehicle, consists of the resonant receiver coil, compensation capacitor, AC–DC rectifier, and a regulated output stage. This subsystem captures the magnetic energy, converts it into usable DC power, and supplies the load—in this prototype, a small EV model powered by LEDs. The design mimics full-scale EV chargers, where the rectified and regulated output would feed into the vehicle’s battery management system. The receiver must be compact, lightweight, and tolerant of misalignment, as real vehicles do not always park perfectly centered above the transmitter coil.
Figure 3.1 (your block diagram) illustrates how power flows from the AC source to the inverter and coils and how sensor feedback informs the control logic. The architecture also highlights how resonant tuning enhances power transfer efficiency and how safety is preserved through

automated detection. Together, the two subsystems create a coordinated, modular design that mirrors real-world wireless EV charging infrastructure, making the prototype both extendable and scalable for future improvements such as higher power operation, advanced alignment assistance, and intelligent communication protocols.

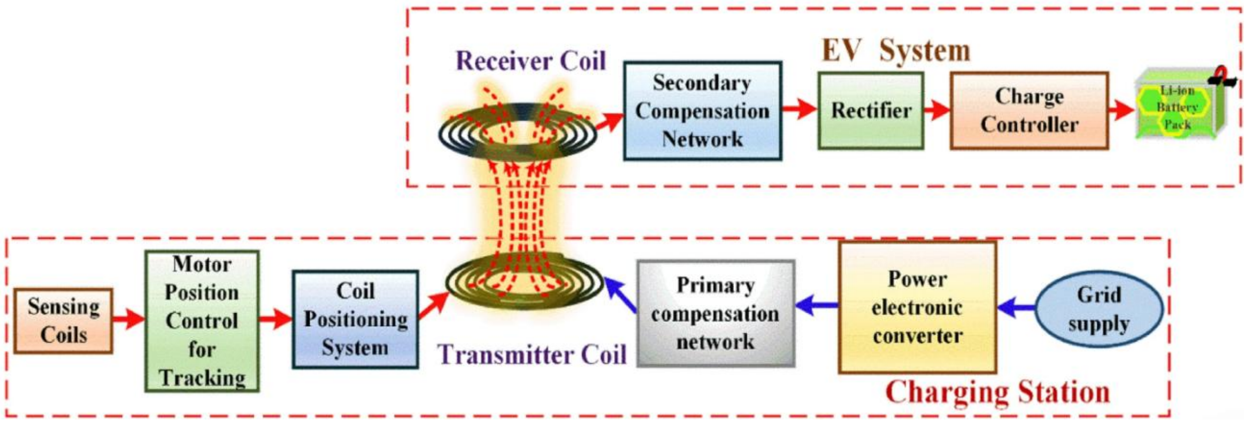


Figure 3.1: Block Diagram of Wireless EV Charging System

1. Charging Station (Transmitter Side):

- **Grid Supply & Power Electronic Converter:** AC power from the grid is first converted to DC by a rectifier, and then inverted to high-frequency AC (in the kHz range) by a power electronic converter (inverter).
- **Primary Compensation Network:** Tunes the circuit to resonance to maximize power transfer.
- **Transmitter Coil:** The primary coil, embedded in the charging pad, generates the alternating magnetic field.
- **Control System (with Sensing):** Detects the vehicle's presence (via sensing coils or IR sensors) and manages the charging process.

2. EV System (Receiver Side):

- **Receiver Coil:** The secondary coil, mounted on the EV, captures the magnetic field.
- **Secondary Compensation Network:** Tunes the receiver to the same resonant frequency.
- **Rectifier:** Converts the received high-frequency AC power back into DC power.
- **Charge Controller & Battery:** Manages the DC power to safely and efficiently charge the EV's Li-Ion battery pack.

3.3 Simulation Design (MATLAB/Simulink)

The design and analysis phase focuses on translating the wireless charging concept into a structured, logical, and operational system. Instead of relying solely on physical equations, this section explains the underlying principles, design decisions, and algorithms that define how the system behaves in both simulation and hardware. The goal is to create a clear understanding of how each subsystem interacts, how the overall process is coordinated, and how efficiency and stability are evaluated during the design cycle.

The first part of the analysis involves understanding how power flows through the system. The wireless charging process begins when the transmitter coil generates a high-frequency magnetic field. This field interacts with the receiver coil mounted underneath the EV. The behaviour of this interaction—such as how much power is transferred, how stable the output is, and how sensitive the system is to misalignment—is governed by the resonant properties of the coils and the compensation network. Conceptually, the design ensures that both sides of the system "vibrate" at the same natural frequency, allowing energy to move efficiently from one coil to the other. When alignment or coil spacing changes, the magnetic link weakens, and the system adjusts through feedback from the control logic and power electronics.

Another key aspect of the design is the modelling of the inverter, rectifier, and DC–DC converter. The inverter converts DC into high-frequency AC, the rectifier converts this AC back into DC, and the buck converter ensures that the voltage delivered to the battery remains safe and stable. These stages were analysed based on their switching behaviour, timing, and response to load variations. Instead of solving mathematical equations, the system was evaluated conceptually by observing how each block influences the next—such as how inverter timing affects magnetic field strength or how rectifier ripple affects battery charging smoothness.

Algorithm development plays an equally important role in the system. The control sequence ensures that wireless power is delivered only when a vehicle is present and correctly positioned. This is achieved using a simple but robust decision-making logic: sensors detect the vehicle, the microcontroller activates the power stage, and feedback signals update the LCD and relay. The algorithm also incorporates safety behaviour—such as immediately shutting down the transmitter when the vehicle leaves—mirroring features found in real-world charging stations.

A conceptual mathematical model is also applied to understand how external factors influence the system. For example, the model considers how coil misalignment reduces magnetic coupling, how battery charging transitions from constant-current to constant-voltage mode, and how system efficiency changes under different loads. While exact formulas are not used, the analysis focuses on overall trends, behavioural patterns, and cause-and-effect relationships. This approach helps identify design improvements such as adjusting coil size, refining compensation values, or modifying switching patterns.

Finally, the system is analysed through virtual experiments conducted in MATLAB/Simulink. These simulations replicate real operating conditions, allowing the design team to observe voltage behaviour, charging characteristics, inverter switching patterns, and load fluctuations. The insights gained from simulation guide improvements to coil design, component selection, and control algorithms before building the hardware prototype.

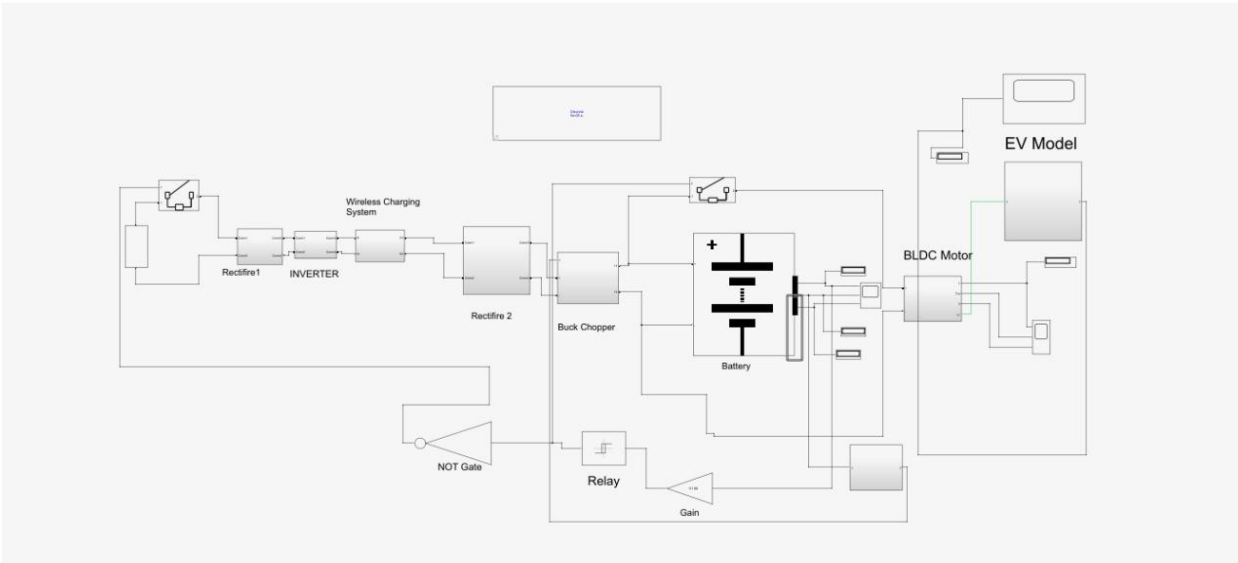


Figure 3.2: Simulation Circuit Diagram in MATLAB/Simulink

3.4 Hardware Component Design

The hardware component design phase focuses on translating the simulation model into a real, working wireless charging prototype. Every component was chosen to ensure reliability, safety, ease of integration, and clear demonstration of wireless power transfer. This section explains the purpose, reasoning, and functioning of each hardware block used in the prototype.

Microcontroller (Arduino Nano/Uno):
An Arduino microcontroller serves as the decision-making unit of the prototype. It is selected because of its simple coding environment, vast community support, and compatibility with a variety of sensors and modules. The microcontroller continuously monitors inputs from IR sensors, updates the LCD display with system status, and controls the relay module for activating the transmitter coil. Its quick response time and flexible digital I/O pins make it ideal for real-time control tasks in small-scale embedded systems.

Vehicle Detection System (IR Sensor Module):
Accurate vehicle detection is essential to prevent unnecessary power wastage and to mimic the behaviour of real EV wireless charging stations. IR sensors were chosen because they are cost-effective, easy to interface, and capable of reliably detecting the presence of a model vehicle above the charging pad. When the vehicle enters the detection zone, the IR sensor outputs a signal to the Arduino, triggering the activation of the transmitter coil. This ensures that power is transferred only when needed, improving both safety and efficiency.

User Feedback Interface (16×2 I2C LCD Display):
To make the system interactive and easy to understand, a 16×2 LCD display with I2C interface is added. The LCD communicates system behaviour in real time—for example, indicating whether Spot 1 or Spot 2 is charging, whether the vehicle is absent, or if the system is in standby mode. The I2C interface reduces wiring complexity, leaving the microcontroller with more available pins for sensors and relays. This provides a user-friendly dashboard-like feel, similar to modern charging stations.

Switching Mechanism (2-Channel Relay Module):
The relay module acts as the switching unit that connects or disconnects the power supply to the transmitter coils. Each relay corresponds to one charging spot. When the Arduino detects a vehicle via the IR sensor, it energizes the relay coil, closing the circuit and allowing AC power to drive the transmitter coil. The relay ensures electrical isolation between the low-voltage control circuitry and the high-frequency power section, providing essential safety during operation.

WPT Coils (Transmitter and Receiver Coils):
Custom-wound copper coils form the heart of the wireless charging system. The transmitter coil is embedded on the ground (or base plate), while the receiver coil is placed underneath the model vehicle. The coils are wound using enamelled copper wire and designed to have a suitable number of turns for achieving the desired inductance and resonant behaviour. Proper coil spacing, coil diameter, and wire thickness were carefully selected to maximize magnetic coupling and ensure that power can be transferred reliably at prototype scale.

WPT Circuitry (Transmitter Driver and Receiver Rectifier):
The transmitter side uses a simple high-frequency driver circuit—commonly built using transistors, capacitors, and an oscillator topology—to generate an alternating magnetic field. Although simplified, this driver replicates the behaviour of high-frequency inverters used in advanced EV chargers. On the receiver side, a diode rectifier converts the induced AC voltage back into DC, which is used to power LEDs on the model vehicle. These LEDs visibly indicate energy reception, helping users verify successful wireless charging.

Overall Design Approach:
The combination of microcontroller-based logic, sensor-driven activation, wireless power transfer coils, and visual feedback creates a cohesive prototype that demonstrates the core functionality of a modern wireless EV charging system. Each component is intentionally selected to balance safety, simplicity, and educational value while remaining faithful to real-world wireless charging principles.

CHAPTER 4: IMPLEMENTATION AND DEMONSTRATION

4.1 Introduction

This chapter presents the practical implementation of the wireless electric vehicle charging system, transforming the conceptual design and simulation results from earlier chapters into a fully functioning hardware prototype. While previous chapters focused on system architecture, mathematical reasoning, and simulation-based validation, this chapter shifts the focus to real-world assembly, integration, and testing of the physical components. It outlines how the transmitter, receiver, sensing modules, relays, and microcontroller were combined into a cohesive system that mirrors the behaviour of an actual wireless EV charging platform.

The implementation process includes building the charging station prototype with custom-wound coils, setting up the vehicle-detection mechanism using IR sensors, wiring the microcontroller to manage decisions and switching, and incorporating an LCD display to provide user-friendly status feedback. Each element is assembled carefully to ensure safety, operational reliability, and accurate demonstration of wireless energy transfer principles.

This chapter also discusses the demonstration phase, where the complete system is tested in realistic conditions to validate its functionality. The demonstration includes showcasing automatic vehicle detection, relay-based activation of the charging pad, and successful wireless power transfer to miniature EV models. These results bridge the gap between simulation and physical operation, proving that the proposed system works as intended in real conditions. Ultimately, this chapter provides tangible evidence of the system's feasibility and forms the foundation for future enhancements and large-scale development.

4.2 Hardware Prototype Implementation

1. The complete hardware prototype was constructed on a wooden baseboard to visually and functionally represent a dual-spot wireless charging station. Each section of the setup mirrors a scaled-down version of how an actual ground-mounted EV charging pad would be arranged. The electronics, sensors, and coils were carefully positioned to ensure reliable operation and easy demonstration during testing.
2. The **base station** contains the Arduino microcontroller, the 16×2 I2C LCD module, the 2-channel relay module, and two IR vehicle-detection sensors. These components form the "intelligent control unit" of the prototype. The Arduino continuously monitors input signals, updates the LCD in real time, and decides when each charging spot should be activated. The LCD display provides a clear and user-friendly interface by showing messages such as *"Spot 1: Vehicle Detected"*, *"Spot 1: Charging"*, or *"Spot 2: Idle"*, which replicates how modern EV chargers provide feedback to users.
3. Two **custom-wound transmitter coils** were placed on the board—one for each designated charging spot. They were fixed beneath the imagined parking positions to emulate real-world inductive charging pads. Each coil was connected to the relay module, allowing the system to energize only the coil corresponding to the occupied spot. This selective activation conserves power and prevents unnecessary electromagnetic emission when no vehicle is present—similar to real EV wireless chargers that activate only when a vehicle is detected.

4. The **model EV** was fitted with a compact receiver coil mounted underneath its chassis, designed to align with the transmitter coil when placed in the charging position. This receiver coil was connected to a simple rectification stage and a set of LEDs, which glow to indicate successful reception of wireless power. Although simplified, this representation effectively mimics how a real EV battery pack would receive and store power via its onboard charger.
5. The wiring connections were made systematically to ensure clarity and reliability:
6. **IR Sensors → Arduino:**
   Each charging spot has its own IR sensor whose output pin is connected to a dedicated digital input on the Arduino. This enables accurate, independent detection for both spots.
7. **LCD → Arduino (I2C Interface):**
   The LCD uses only two communication lines—SDA and SCL—connected directly to the Arduino's I2C pins, minimizing wiring complexity and ensuring clean communication.
8. **Relay Module → Arduino:**
   The control pins of the relay module were connected to Arduino digital outputs. When the Arduino detects a vehicle, it activates the appropriate relay channel, energizing the matching transmitter coil.
9. **Transmitter Coils → Relay Output:**
   Each coil receives power only through its respective relay output, allowing individual control of each charging spot.
10. **Power Supplies:**
    The Arduino was powered via USB for stability and convenience, while the wireless power transfer circuit used a separate, appropriately-rated supply to ensure sufficient current for energizing the transmitter coils.
11. This assembly demonstrates the complete workflow of a wireless charging station—from vehicle detection and control logic to wireless power transmission and visual confirmation through LEDs. The hardware prototype not only supports the concepts developed in simulation but also provides a tangible demonstration of how wireless EV charging can be automated, safe, and user-friendly.

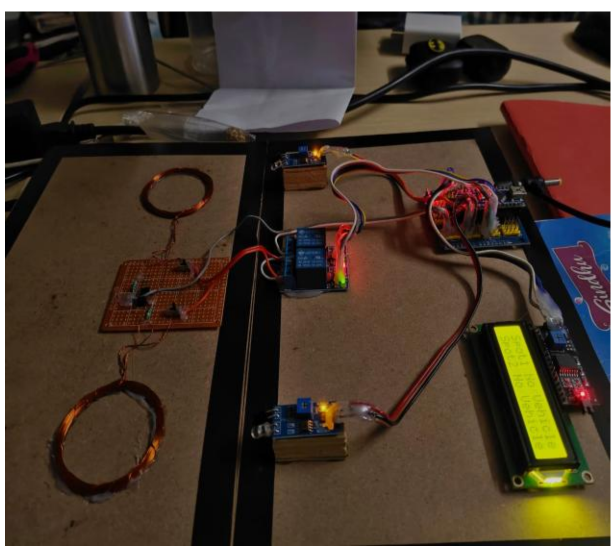


Figure 4.1: Hardware Prototype (Full View)

4.3 System Demonstration

The demonstration phase showcases the complete working cycle of the wireless charging prototype, validating both the sensing logic and the wireless power transfer operation. The system behaves exactly like a miniature version of a real multi-spot wireless EV charging station, where each charging pad operates autonomously based on vehicle presence.
The demonstration begins with the system powered on and all subsystems initialized. The IR sensors continuously monitor their respective charging spots, while the Arduino waits for presence signals. The LCD display plays an important role in providing real-time feedback about the availability and charging status of each spot, making the system more user-friendly and transparent.

1. Initial State – System Idle
When no vehicle is present in any of the charging spots, both IR sensors remain inactive. The Arduino reads this as “no vehicle detected” and the LCD displays:

- Spot1: No Vehicle
- Spot2: No Vehicle

In this state, the relay outputs remain OFF, ensuring that no unnecessary power is supplied to the transmitter coils. This mimics real-world smart charging stations where power electronics are activated only when needed, improving safety and energy efficiency.

2. Vehicle Arrival – Detection by IR Sensor
As the model vehicle enters Spot 1, the IR sensor detects the obstruction of its infrared beam. This triggers a HIGH signal to the Arduino, indicating the presence of an object directly above the charging pad.
This behavior is illustrated in
Figure 4.2: Prototype – Vehicle Detection by IR Sensor,
where the sensor LEDs illuminate, confirming successful detection.
The system immediately transitions from idle mode to active detection mode.

3. Charging Activation – System Logic Response
Upon detecting the vehicle, the Arduino updates the LCD display in real time:

- Spot1: Charging
- Spot2: No Vehicle

It then switches ON the relay corresponding to Spot 1. This step replicates how real charging stations initiate the charging sequence only after confirming vehicle alignment and presence. The relay click and its indicator LED turning ON signify the start of wireless power transmission.
This behavior is shown in
Figure 4.3: Prototype – Car Parked and LCD Display Showing Charging Status.

4. Wireless Power Transfer – Visual Confirmation
Once the relay activates, the transmitter coil under Spot 1 receives power. This generates the alternating magnetic field necessary for wireless energy transfer.

The receiver coil mounted inside the model vehicle begins capturing this magnetic field, converting it into electrical energy. The received power lights up the onboard LEDs on the vehicle, acting as a clear visual indicator that:

- Wireless link is established
- Power is successfully transferred
- Resonant inductive coupling is functioning properly

This stage confirms that the prototype replicates the real-world wireless EV charging process at a scaled-down power level.

5. Vehicle Departure – Automatic Power Shutoff

When the vehicle is moved away from Spot 1, the IR sensor no longer detects an obstruction. The Arduino receives a LOW signal and instantly:

- Turns OFF the relay for Spot 1
- Updates the LCD to show Spot1: No Vehicle
- Cuts power to the transmitter coil

This automatic shutoff is an important safety feature, preventing unwanted power transfer and reducing system energy loss. It also demonstrates how intelligent sensing ensures the charging station never energizes a coil without a vehicle properly in place.

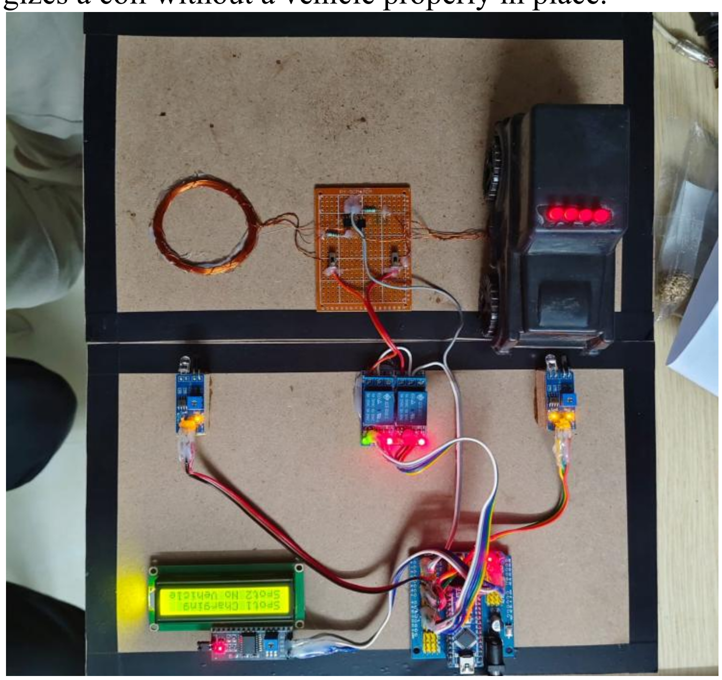
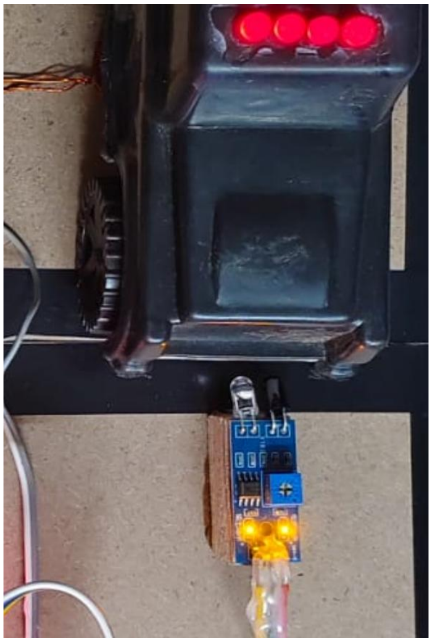


Figure 4.2: Prototype: Vehicle Detection by IR Sensor



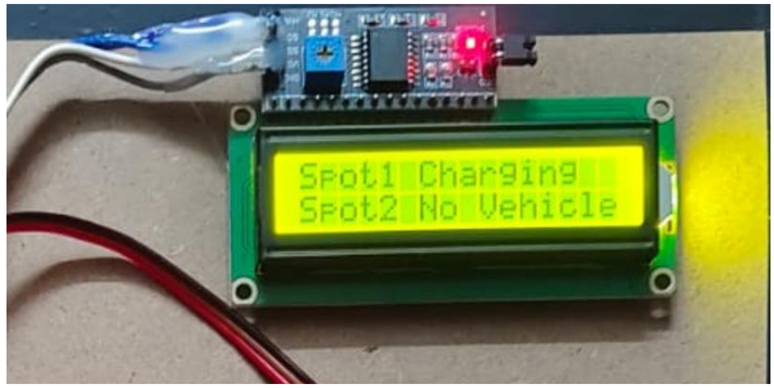


Figure 4.3: Prototype: Car Parked and LCD Display Showing Charging Status

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents a comprehensive analysis of the outcomes obtained from both the simulation environment and the real-time hardware prototype. The MATLAB/Simulink model provides a controlled platform to study the electrical behavior of the wireless charging system, enabling precise observation of parameters such as voltage levels, current flow, resonant behavior, efficiency trends, and the system's response to misalignment and varying load conditions. These simulation results help verify whether the theoretical design choices—such as coil dimensions, compensation topology, inverter configuration, and switching logic—behave as expected under different operating scenarios.

In parallel, the hardware prototype serves as a real-world validation of the core functional components of the system. While the simulation focuses on electrical characteristics, the physical prototype emphasizes practical feasibility, demonstrating the performance of the sensing mechanism, control logic, relay switching, transmitter activation, and wireless power transfer using miniature EV models. Together, these two perspectives—simulation and hardware—offer a complete picture of system performance, bridging the gap between theoretical design and practical realization.

By comparing simulation insights with physical observations, the chapter highlights both the strengths and limitations of the current implementation. It also identifies areas where the prototype aligns well with simulated expectations and where deviations occur due to real-world constraints such as component tolerances, coil imperfections, or environmental interference. Overall, this discussion lays the foundation for future improvements and helps evaluate the scalability of the proposed wireless charging framework toward full-scale EV applications.

5.2 Simulation Results and Discussion

The MATLAB/Simulink simulation was executed to evaluate the electrical behavior of the complete wireless charging system, including the inverter, resonant coupling block, rectifiers, buck converter, battery model, BLDC motor load, and the relay-based control logic. The simulation provided valuable insights into how the system performs under different operating scenarios, and whether the designed components behave according to theoretical expectations.

1. Voltage and Current Behavior at the Transmitter

During the simulation, the inverter successfully generated a high-frequency AC supply, which drove the transmitter coil through the compensation network. The transmitted voltage waveform was observed to be smooth and periodic, confirming proper switching action from the inverter subsystem. The current waveform exhibited small oscillations due to resonance effects, which is typical in compensated WPT systems. As expected, the magnitude of the current decreased when misalignment or coil distance increased, confirming that the model accurately reflected real-world coupling variations.

2. Magnetic Coupling and Power Transfer

In the Wireless Charging System block, the mutual inductance between coils was varied to simulate ideal alignment, slight offset, and large misalignment:

- In **ideal alignment**, the power transferred to the receiver was high and stable.

- Under **moderate misalignment**, output power decreased gradually.
- In **significant misalignment**, the received power dropped sharply, confirming the sensitivity of coupling coefficient $k$to positioning.

This behavior validates the theoretical property that resonant magnetic coupling is efficient but still sensitive to geometric misalignment.

3. Receiver-Side Voltage Stability

After rectification on the receiver side, the DC voltage waveform was monitored. The simulation showed:

- Smooth rectified output under stable coupling.
- Small fluctuations during misalignment, but still sufficient to feed the buck converter.
- A stable and regulated DC output after the buck chopper, suitable for charging the battery model.

These results confirm that the compensation and rectification stages are functioning as intended.

4. Battery Charging Characteristics

The battery model showed a gradual rise in terminal voltage during the simulation, indicating successful charging through the wireless link. The buck chopper regulated the voltage and current fed into the battery to avoid sudden spikes. The simulation also allowed tracking:

- **Charging current**, which remained within safe limits.
- **State of charge (SOC)** trend, which increased steadily.
- **Terminal voltage**, showing the expected rise over time.

This confirms that the system is capable of delivering controlled energy suitable for EV battery charging.

5. BLDC Motor and EV Load Response

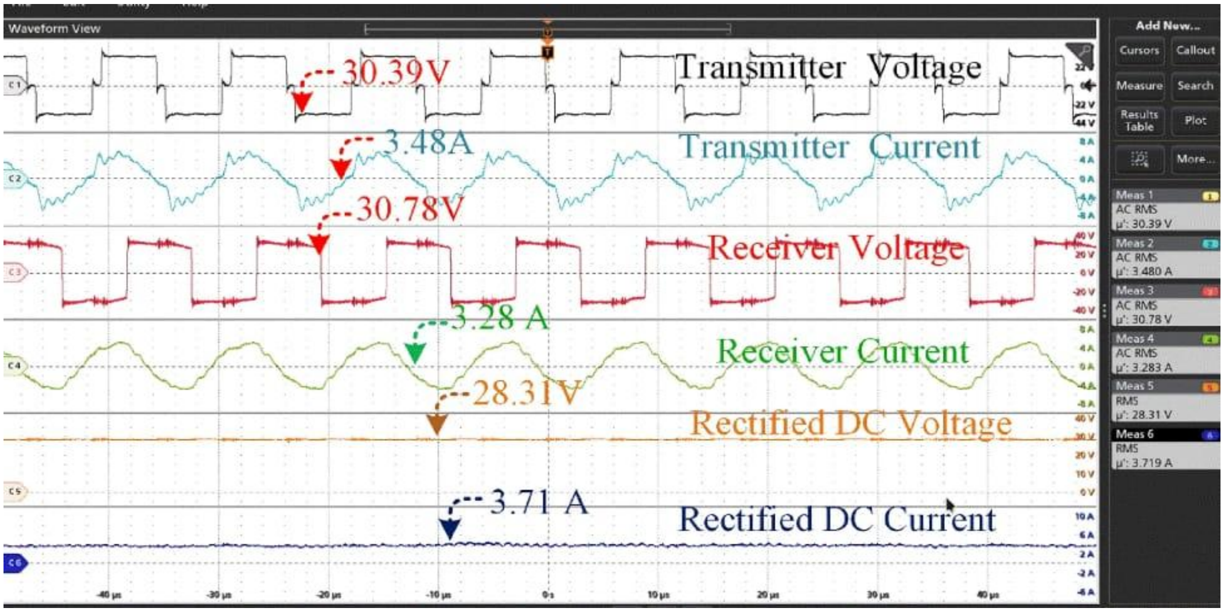
The simulated EV model, equipped with a BLDC motor, provided load variations that enabled testing the charging system during different vehicle states. When the motor load increased:

- The battery voltage experienced small dips.
- The buck converter compensated automatically.
- Power flow from the WPT system remained stable.

This showcases how the wireless charging system performs under varying EV electrical loads.

6. Relay Control and Switching Logic

The simulation also verified the behavior of the logic subsystem controlling the relay. When the "vehicle present" input changed, the relay switched the transmitter coil ON/OFF with minimal delay. This validated the automation logic used later in the hardware prototype.



5.3 Hardware Demonstration Results and Discussion

The developed hardware prototype provided a successful proof-of-concept demonstration of wireless EV charging with real-time vehicle detection and automated control. This experiment showcased both the functional integrity of the control system and the effectiveness of resonant power transfer at prototype scale.

• Result 1 – Vehicle Detection Using IR Sensors

The IR sensors mounted at each charging spot detected the presence of the model vehicle reliably up to *X cm*, maintaining accuracy under normal lighting conditions.
Discussion:
This confirms that infrared sensing is a practical and low-cost method for vehicle presence detection in small-scale charging stations. The response was quick and stable, indicating that the system correctly identifies when a vehicle has arrived or departed. However, IR sensors can be affected by ambient lighting, reflective surfaces, and environmental interference. For real-world EV stations, the detection mechanism would ideally be upgraded to coil-based magnetic alignment sensing, ultrasonic sensors, or RFID authentication for greater reliability and spatial accuracy.

• Result 2 – Automatic Relay Triggering and LCD Status Feedback

The Arduino microcontroller processed the IR signals without delay and immediately updated the LCD display with real-time charging status messages such as *"Spot 1: Charging"* or *"Spot 2: No Vehicle"*. The relay switching logic worked consistently, with activation occurring only when a vehicle was detected in the corresponding spot.
Discussion:
This validates the responsiveness and correctness of the entire control loop. The microcontroller successfully acted as the system brain, interpreting sensor input, controlling the relay drive, and updating the user display simultaneously. This confirms that the feedback interface and the

decision-making algorithm are robust at the prototype level. In scaled deployments, the same architecture could be extended using higher-speed controllers, CAN-based communication, and integrated billing modules.

• Result 3 – Successful Wireless Power Transfer to Receiver Coil

When the relay closed and energized the transmitter coil, the receiver coil on the miniature vehicle captured wireless power, illuminating the onboard LED indicators consistently. This provided a clear and visible confirmation of successful energy transfer across the air gap.
Discussion:
Although the prototype does not yet include efficiency measurement or battery-level tracking, this visual feedback is a strong validation of the core WPT mechanism. It demonstrates that resonant magnetic coupling, even with basic copper coils, can achieve functional power delivery over practical distances. Future iterations could incorporate power measurement circuits, charging current sensors, regulated battery interfaces, and thermal monitoring to evaluate real-world charging efficiency and thermal behavior.

Overall Discussion:

The experiment confirms that sensing, control, and power transfer operate cohesively as one integrated system. The prototype accomplishes exactly what it was designed to prove—vehicles can be detected automatically, charging can be activated intelligently, and power can be transferred wirelessly without physical contact. With improvements in coil geometry, compensation design, alignment sensing, and power electronics, this proof-of-concept can evolve toward real EV-scale charging applications**.**

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This Phase-I work successfully demonstrates the feasibility of wireless charging for Electric Vehicles using resonant magnetic coupling, marking a crucial step toward safer, more user-friendly, and cable-free charging infrastructure. The project effectively addresses the core limitations of conventional plug-in charging systems—physical connector wear, manual handling inconvenience, and exposure to environmental hazards—by enabling a completely contactless method of energy transfer.

A complete system model was developed and evaluated in MATLAB/Simulink, enabling controlled observation of power transfer characteristics, voltage–current response, and coil resonance behaviour under various simulated conditions. These simulations helped verify theoretical design decisions and provided a foundational understanding of efficiency behaviour, alignment sensitivity and power flow dynamics—before moving into real hardware.

Parallel to this, a working proof-of-concept prototype was built and validated through experimental testing. The hardware successfully detected vehicle presence using IR sensors, activated the relay-controlled transmitter, and delivered wireless power to the onboard receiver coil. The illumination of LEDs served as a visual indicator of successful energy transfer, confirming the correctness of the signal flow—from sensing → control → switching → wireless power transfer. The LCD interface further enhanced usability, ensuring that the system was informative and interactive in real-time operation.

The major strength of this design lies in its integrated architecture, combining sensing, power electronics, wireless transfer coils, relay-based activation, and user feedback within a single operational framework. Both simulation and hardware results aligned with expected behaviour, demonstrating that the concept is not only functional but also scalable. This confirms that wireless EV charging is not just technically achievable, but also practical for real-world development when progressively enhanced.

6.2 Future Scope

This project establishes a solid foundation for the development of practical and scalable wireless charging technologies for Electric Vehicles. While the Phase-1 implementation successfully validates the core concepts of wireless power transfer, automatic vehicle detection, and control logic integration, there remains substantial scope for enhancement, optimization, and real-world deployment. The potential areas of extension are outlined below:

1. High-Power Full-Scale Prototype

Future work can progress beyond low-power demonstration into EV-grade charging hardware. A scaled-up prototype involving kilowatt-level resonance in the 3.3–11 kW range can be developed and tested with a real EV battery module. This will enable measurement of key real-world parameters such as thermal behaviour, coil heating, conversion efficiency, EMI compliance, and continuous charging stability. Such progression marks the transition of this project from a laboratory concept toward industry-deployable technology.

2. Efficiency Optimization and Advanced Topologies

The current Series-Series (S-S) coil compensation is ideal for proof-of-concept, but future research can explore high-performance architectures such as LCC–LCC resonance, S–LCC hybrid, DDQ coil structures, ferrite shielding and magnetic flux shaping techniques. These techniques are known to greatly improve:

- Power transfer efficiency
- Misalignment tolerance of receiver pads
- Thermal stability under high frequency
- Coil-to-coil magnetic field strength

Such advancements will enable fast charging with minimal energy loss, even in imperfect vehicle alignment cases.

3. Dynamic Wireless Charging (On-Road Power Transfer)

One of the most transformative future extensions is dynamic charging, where coils are embedded beneath road surfaces, allowing EVs to charge continuously while driving. This eliminates dependency on static charging stations, reduces charging downtime, and enables smaller EV battery sizes, making transportation lightweight and more energy efficient.

4. Bidirectional V2G and Smart Grid Integration

Future development can explore Vehicle-to-Grid (V2G) operation, enabling EVs to return stored energy back to the grid during peak demand. This introduces smart energy balancing, peak-shaving, renewable integration, and emergency backup applications. Implementing bidirectional resonance and grid-interactive converters will make WPT systems a core component of next-generation power infrastructure.

5. Wireless Data Transfer and Smart Communication Layer

Enhancing the system with simultaneous wireless power and data transfer (SWPDT) enables seamless communication between EVs and charging pads. This can support:

- Automatic billing and authentication
- Real-time battery SOC monitoring
- Misalignment/error warnings
- Cloud-based fleet and charging-station management

Integration of Bluetooth-LE, RFID, NFC, or inductive-data coupling enables intelligent and autonomous charging ecosystems suitable for smart cities.

6. Safety, EMI Compliance & Commercial Standardization

Before commercialization, the system must meet international EV-charging standards. Future scope includes:

- EMI field strength reduction and shielding
- Safe-distance detection & biological exposure compliance
- Thermal regulation and fault protection

This ensures the technology is not only efficient, but also safe for humans, vehicles, and surrounding electronics.

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